Study Overview

This study proposes a hybrid MARUS–PINN framework for identifying ship manoeuvring hydrodynamic derivatives. The framework integrates the MARUS simulator, a Unity-based real-time marine simulation platform, with a Physics-Informed Neural Network (PINN) and symbolic regression. The aim is to evaluate whether this workflow can generate hydrodynamic models of autonomous surface vessels (ASVs) that are physically interpretable, accurate, and generalisable, offering improved fidelity and reliability compared to conventional system identification or purely data-driven methods.
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1. Simulation and Data Generation

A comprehensive set of standard IMO/ITTC manoeuvres—Turning Circle, Zig–Zag (10/10° and 20/20°), Acceleration/Deceleration, and PRBS Rudder excitation—was performed within the MARUS simulator environment using predefined control routines implemented through dedicated C# manoeuvre scripts. These manoeuvres were designed to excite the vessel's primary dynamic modes in surge, sway, and yaw, enabling estimation of classical MMG-type hydrodynamic derivatives such as $Y_v, N_r, N_\delta, X_u, X_{uu}$.

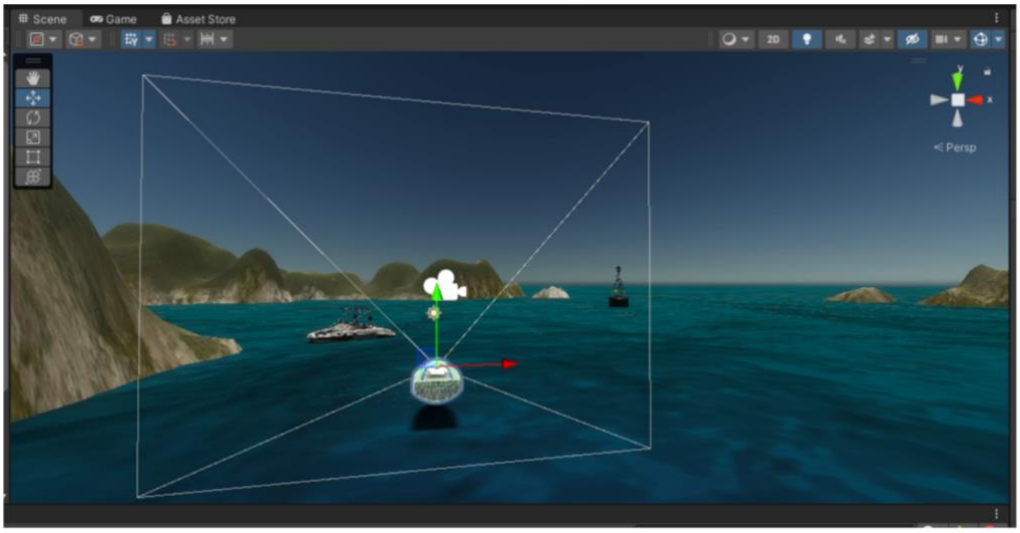

Figure 1 Overview of MARUS example scene.

Maneuver	Primary derivatives excited
Turning Circle (TC)	Y_v, N_r, N_δ
Zig–Zag (ZZ 10/10°, 20/20°)	Y_δ, N_δ (control response)
Acceleration / Deceleration / Crash-Stop	X_u, X_{uu}
PRBS Rudder	Coupled derivatives via response spectrum

Figure 2 Standard manoeuvres and primary hydrodynamic derivatives excited.

MARUS, developed on the Unity engine, was selected over alternative platforms such as STAR-CCM+, OpenFOAM, and ASVSim because it offers an effective balannce between runtime performance, openness, and integration readiness. While high-fidelity CFD tools like STAR-CCM+ and OpenFOAM provide benchmark accuracy, they remain computationally prohibitive for real-time or closed-loop applications. In contrast, MARUS delivers real-time marine simulation with local ROS2 and gRPC connectivity, enabling seamless middleware integration, dataset generation, and autonomy testing. Its Unity-based architecture supports structured logging and live data exchange, making it particularly suitable for System Identification and Digital Twin frameworks. Although MARUS uses empirical hydrodynamic models rather than fully resolved CFD, its DT/SI-ready interfaces make it an appropriate platform for hybrid grey-box frameworks such as the proposed MMG + PINN model in this study.

All manoeuvres are executed autonomously under Unity's fixed-step physics loop to ensure error-free timing updates. The unified control and logging architecture, using essential functionality from the ControlBus and BoatDataLogger scripts explained in the subsequent parts, ensures that each input–output pair reflects the exact state of the vessel at every simulation run. Across all manoeuvre types (Turning Circle, Zig–Zag , Acceleration/Deceleration, Crash-Stop, and PRBS Rudder) a total of 36 CSV datasets were generated, covering three throttle configurations (0.6, 0.8, and 1.0) to capture low-, medium-, and high-speed hydrodynamic modes. Each dataset contains global position, unwrapped heading, body-fixed velocities (u, v, r), differential and total thrust inputs, torque, and rudder-proxy signals. Together, this structured dataset provides comprehensive excitation of the vessel's dynamic modes, forming a consistent empirical foundation for estimating classical MMG-type hydrodynamic derivatives and for training and validating the hybrid MMG + PINN residual-learning framework elaborated in the following sections.

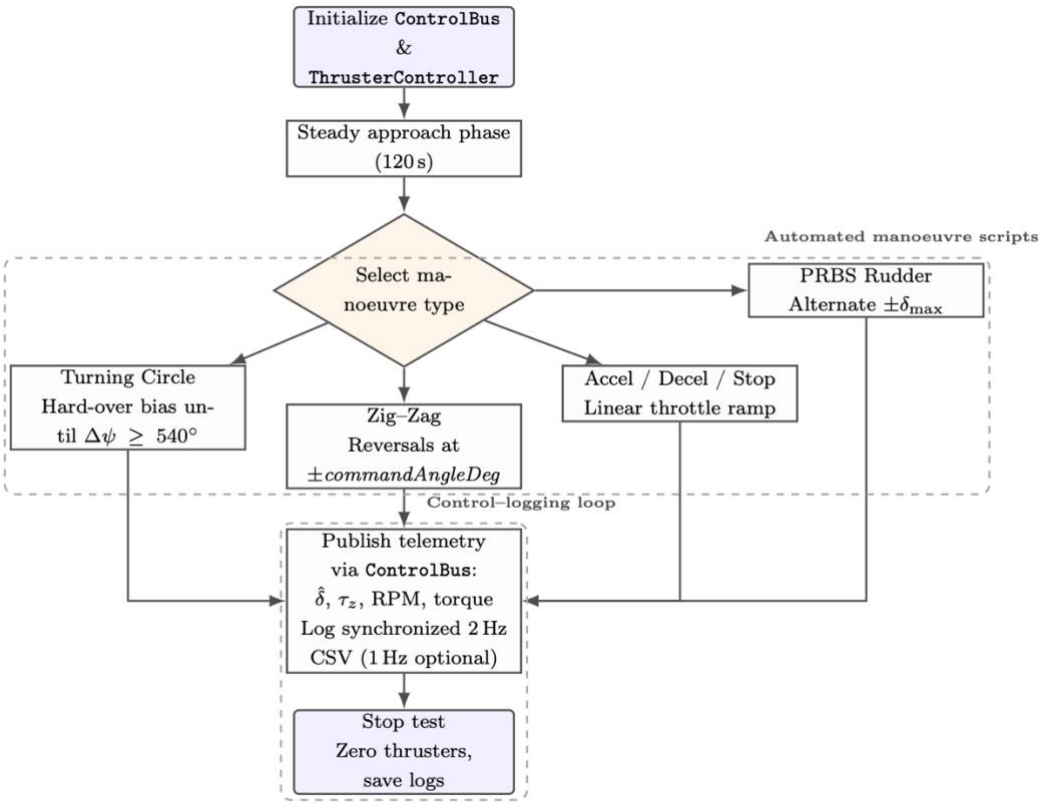

Figure 3 Structured flow of the control–logging framework in MARUS.

2. Data Preprocessing and Physical Labeling

All simulation data were processed using a modular Python-based preprocessing pipeline.

This included:

- Timing audits for drift and jitter correction.
- Resampling to a 2 Hz time base.
- Low-pass filtering (third-order Butterworth, 0.5 Hz cutoff) to remove noise.
- Outlier correction using Z-score clipping and rolling median–MAD imputation.
- Computation of derived features such as resultant speed and normalized rudder inputs.

Additional metrics (such as Advance, Transfer, Overshoot Angle, and Stopping Distance) were extracted for manoeuvre-specific quality control. The resulting dataset was standardized for system identification and PINN training. The entire preprocessing pipeline can be seen below:

3. Grey-Box PINN Architecture

The proposed Grey-Box PINN integrates a fully differentiable MMG-based dynamic core within its neural computation code, providing an explicit physical model as a prior over surge, sway, and yaw dynamics. Unlike conventional black-box PINNs, the model enforces physically meaningful parameter constraints and learns only residual corrections beyond the baseline MMG model.

Key architectural contributions:

- **Soft-constrained hydrodynamic parameters**: positive inertia and added-mass terms, and

Input: Raw MARUS CSVs $\mathcal{F} = \{f_i\}$; sampling rate $f_s = 2\,\text{Hz}$; cutoff $f_c = 0.5\,\text{Hz}$
Output: Clean, validated dataset $\mathcal{D}$ with manoeuvre metrics, identifiability, uncertainty, and FAIR metadata

function PREPROCESSDATASET($\mathcal{F}$)
  Init empty lists for processed frames, metrics, audits, metadata.
  for each $f_i \in \mathcal{F}$ **do**
    Load CSV; inspect stats and Δt stability.
    Run timing audit (jitter, drift_{ppm}, missing rate).
    Uniformly resample $\Delta t{=}1/f_s$; interpolate gaps.
    Low-pass filter (u, v, r) via Butterworth(f_c,3); z-score clip outliers.
    Unwrap heading; compute $r_{\text{heading}}{=}d\psi/dt$; store $(u_f, v_f, r_f, \delta, U)$.
0.3em // Outlier Cleaning and Yaw Alignment
  for each D_i **do**
    Apply rolling median+MAD filter; replace spikes.
    Bandlimit and cross-correlate r_f vs. r_{heading} to find lag.
    Shift, interpolate, and flag "ok/suspicious" via RMSE test.
0.3em // Manoeuvre Classification and Metrics
  for each D_i **do**
    Detect type $\in$ {TC, ZZ10, ZZ20, PRBS, ACCEL, DECEL, STOP}.
    if TC **then** Compute Advance, Transfer, Tactical Diam. ($/L_{PP}$).
    else if ZZ **then** Derive 1st/2nd overshoots; detect reversals.
    else if PRBS **then** Detect steps; compute PSD energy, peak freq, coherence $\gamma^2_{r\delta}(f)$.
    else if ACCEL/DECEL/STOP **then** Derive time-to-90%, mean accel/decel, or stopping distance.
0.3em // Identifiability & Uncertainty
  for each D_i **do**
    Build $\Phi{=}[r, |r|r, \delta]$; compute $\kappa(\Phi^\top\Phi)$.
    Compute $\rho_{r,\delta}$; flag if $\kappa{<}10^8$ & $|\rho|{>}0.5$.
    Estimate $R{=}U/|r|$; bootstrap $\text{CI}_{95\%}$.
0.3em // Surge Dynamics
  for ACCEL, DECEL **do**
    Fit $u{=}u_\infty(1 - e^{-t/\tau_a})$ or $u{=}u_0 e^{-t/\tau_d}$; extract τ.
0.3em // Nomoto 1st-Order Model
  for TC, ZZ, PRBS **do**
    Fit $T\dot{r} + r = K\delta$ via ridge LSQ; save (K, T, σ_{res}).
0.3em // FAIR Output Assembly
  Aggregate all runs $\rightarrow \mathcal{D}$; export `pinn_ready` train/val/test CSVs.
  Generate dataset card PDF.
0.3em **return** FAIR-compliant data.

negative damping coefficients, imposed via differentiable activations such as softplus and sigmoid to guarantee physical plausibility.

- **Rudder-moment sign enforcement**: ensures the learned yaw-moment coefficient retains the correct physical directionality
- **Residual neural block in physical units**: capturing unmodelled nonlinearities such as rudder–hull and cross-flow coupling without offsetting the physical scaling.
- **Roll-out stability regularization**: a long-horizon penalty that constrains unstable divergence, improving simulation fidelity under free-running marine conditions.

Together these components deliver a model that is both interpretable and data-adaptive, forming a hybrid physics–learning framework that preserves the traditional MMG structure while enabling data-driven refinement of higher-order hydrodynamic effects.

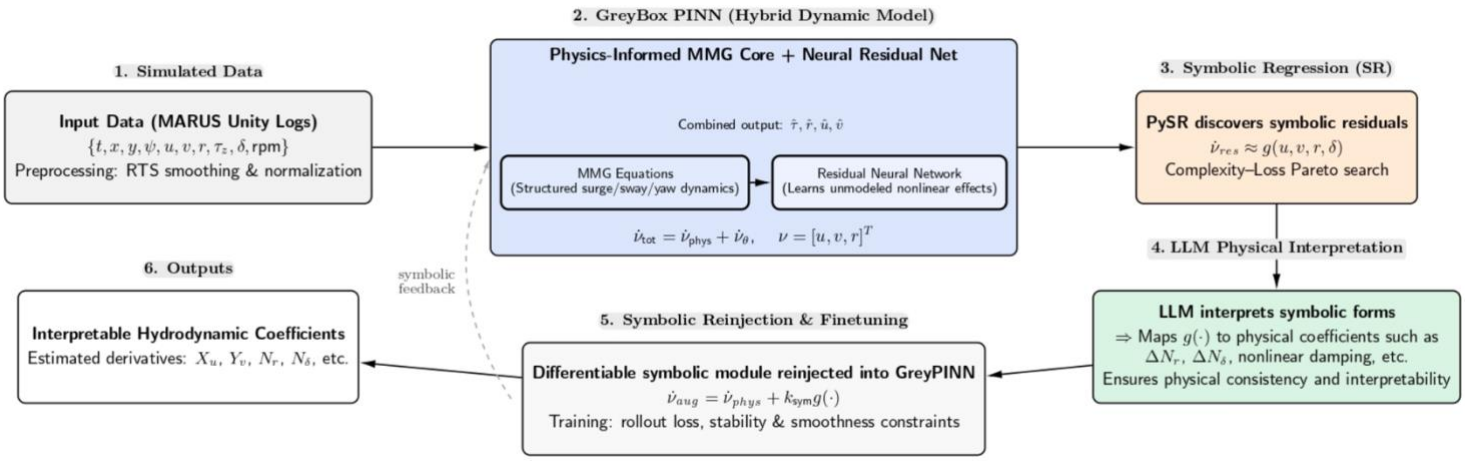

Figure 4 Overview of the proposed hybrid GreyBox PINN workflow.

3. Symbolic Regression and LLM-Assisted Interpretation

To uncover physically meaningful patterns within the learned residuals, the post-training stage applies multi-channel symbolic regression (via PySR) to each dynamic equation (surge, sway, yaw). This results in compact numerical expressions that expose hidden nonlinearities and coupling effects (such as asymmetric yaw damping or rudder–hull interference) that are otherwise unobservable in neural weights.

An LLM-based hydrodynamic interpreter subsequently classifies these symbolic terms into standard MMG notation and generates structured explanations. This automated translation bridges AI-based corrections with the literature-established hydrodynamic terminology, turning the residual model into an interpretable extension of MMG theory.

This symbolic–LLM coupling is a novel interpretability layer within the hybrid framework, allowing empirical discovery of new hydrodynamic relationships while maintaining a physically accurate representation.

Input: Normalized symbolic expression $f(\mathbf{x})$, scaling parameters (μ, σ), vessel context

1: Convert to physical form $f_{\text{phys}}(u_f, v_f, r_f, \delta)$ using μ, σ

2: Construct structured prompt:

- Role: `"You are a naval hydrodynamics researcher"`
- Input: normalized + physical expression
- Context: vessel type, speed, rudder limits
- Task: identify MMG-style terms (N'_r, N'_δ, Y'_δ)
- Output: JSON {effects, description, confidence, interpretation}

3: Query LLM (`Gemini 2.5 Pro`) $\rightarrow$ receive raw JSON output

4: Parse effects $\rightarrow$ store as `symbolic_interpretation_yaw.json`

5: Second-stage prompt: rewrite JSON summary in academic hydrodynamic language

Output: Final structured output:

$$\text{Effects: } \{(\text{name}, \text{description}, c_i)\},$$
$$\text{Interpretation: Hydrodynamic paragraph (100–150 words)}$$

Figure 5 Prompt-engineering workflow for physics-aware LLM interpretation of symbolic residuals.

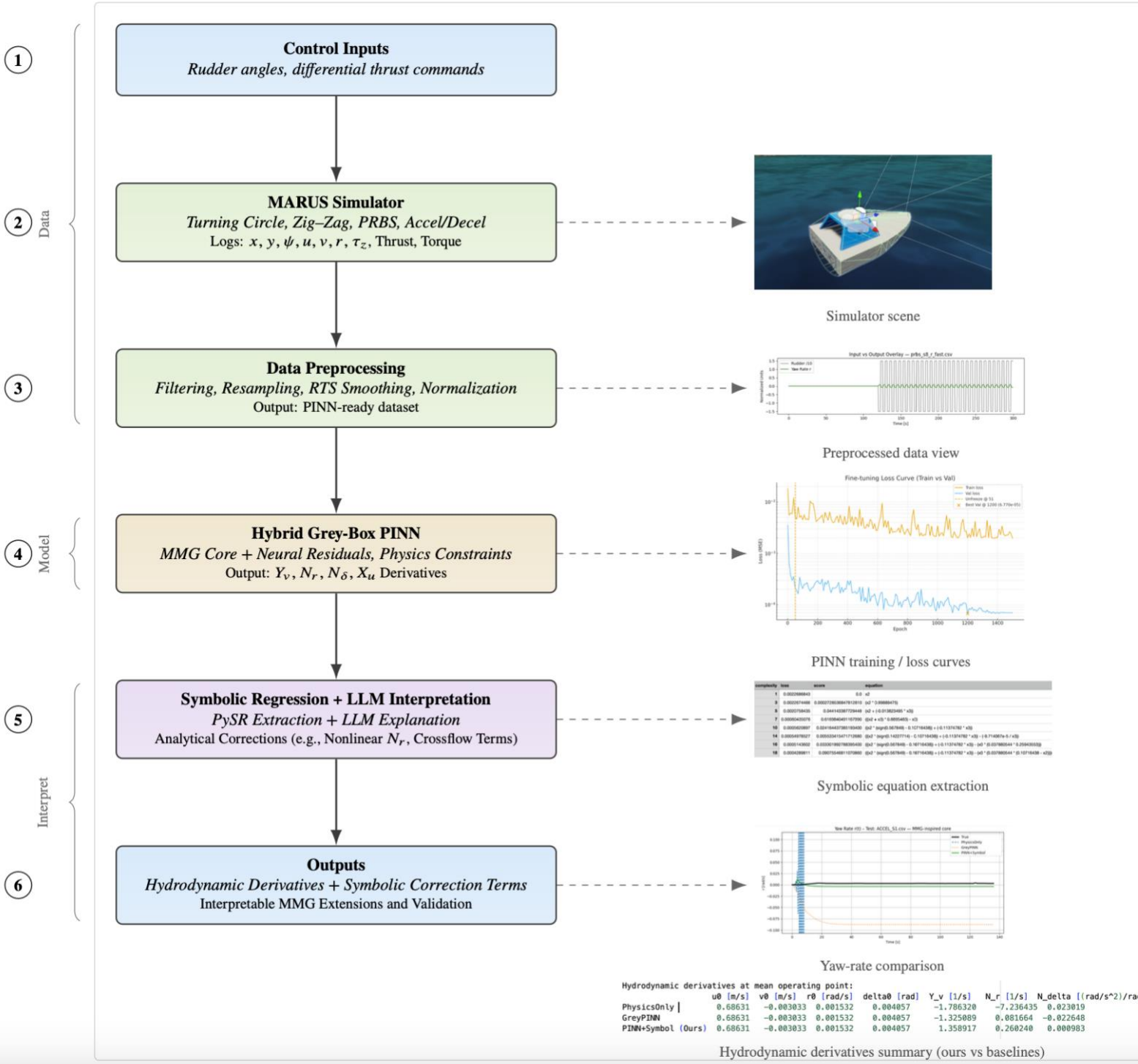

Figure 6: Hybrid MARUS–PINN Framework with LLM-Guided Symbolic Discovery

4. Model Evaluation and Benchmarking

To evaluate model **generalizability and physical consistency**, three variants were tested on an independent MARUS dataset **outside the training domain: PhysicsOnly** (baseline; pure MMG equations without learning), **GreyPINN** (baseline; MMG equations augmented with a small neural residual term), and **PINN + Symbol** (final hybrid model; PINN with post-hoc symbolic correction and interpretability layer).
Acceleration and rollout RMSE metrics were computed for surge, sway, and yaw variables. The **results**, show mean RMSE values for surge (u), sway (v), yaw-rate (r), and rollout errors in r and heading ψ. The **PINN + Symbol model achieves the lowest errors across all variables**, indicating superior generalization and long-horizon stability.

Model	$\mathbf{RMSE}_u$ **[m/s]**	$\mathbf{RMSE}_v$ **[m/s]**	$\mathbf{RMSE}_r$ **[rad/s]**	**Rollout** $\mathbf{RMSE}_r$ **[rad/s]**	$\mathbf{RMSE}_\psi$ **[rad]**
PhysicsOnly	0.0929 ± 0.0005	0.0274 ± 0.00002	0.1779 ± 0.0002	9999 ± 0	9999 ± 0
GreyPINN	0.0130 ± 0.0002	0.0154 ± 0.0072	0.0191 ± 0.0006	$476.19^{+880.86}_{-476.15}$	$480.98^{+880.89}_{-477.18}$
PINN+Symbol	$\mathbf{0.0130 \pm 0.0001}$	$\mathbf{0.0039 \pm 0.0009}$	$\mathbf{0.0056 \pm 0.0009}$	$\mathbf{0.0312^{+0.0042}_{-0.0076}}$	$\mathbf{1.443^{+0.607}_{-0.568}}$

Figure 7: Final Combined Benchmark Report

Summery

This study introduces a hybrid grey-box learning system that merges MARUS simulation, data-driven residual learning, and symbolic–LLM interpretability. By unifying physics-based modelling with modern AI, it offers a reproducible, interpretable, and high-fidelity approach for hydrodynamic system identification and digital twin development in marine robotics.